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Course: CSA0421 - Operating System **GitHub Repo:** krishshiva721/CSA04--Assignment1

TASK 1: REQUIREMENTS EVALUATION

Evaluate and Justify Operating-System Design for a Smart Hospital

1. COMPREHENSIVE COMPUTING ENVIRONMENT ANALYSIS

A 500-bed smart hospital encompasses distinct operational domains. Each device type presents conflicting operational metrics across five core pillars: Response Time, Reliability, Throughput, Battery/Power Efficiency, and Security.

Environment	Primary Constraint	Key Metrics	Justification & Requirement Ranking
Bedside Patient Monitors	Critical Safety	Response: < 10 ms Reliability: 99.999%	**Rank: 1. Response Time, 2. Reliability, 3. Security, 4. Throughput, 5. Battery.** Deterministic execution is mandatory. Late alarm signals risk patient death; throughput and UI complexity are secondary.
Nurses' Handheld Tablets	Mobile Usability	Battery: > 12 hrs Security: HIPAA / Role-based	**Rank: 1. Battery Life, 2. Security, 3. Reliability, 4. Response Time, 5. Throughput.** Shift survival (12+ hrs) without thermal/power exhaustion is critical alongside protecting mobile PHI data.
Doctors' Workstations	Interactive UX	Response: < 500 ms Throughput: Multi-app	**Rank: 1. Reliability, 2. Response Time, 3. Throughput, 4. Security, 5. Battery.** Desktop workstations require fast interactive response for EHR input and multi-window multitasking without system crashes.
Central Database Server	Data Availability	Reliability: Zero Downtime Throughput: High IOPS	**Rank: 1. Reliability, 2. Security, 3. Throughput, 4. Response Time, 5. Battery.** Patient medical records must remain available 24/7. System halts or uncommitted write failures compromise overall care.
MRI Imaging Server	Compute Intensive	Throughput: Multi-GB/s CPU Utilization: High	**Rank: 1. Throughput, 2. Reliability, 3. Security, 4. Response Time, 5. Battery.** Heavy computational workloads require maximum parallel execution over multi-core/GPU architectures.
Partner Hospital Link	Interoperability	Security: TLS / Encryption Reliability: Packet Delivery	**Rank: 1. Security, 2. Reliability, 3. Throughput, 4. Response Time, 5. Battery.** Boundary protection, end-to-end encryption, and federated identity verification take absolute precedence.

2. REQUIREMENT CONFLICTS & HISTORICAL OS EVOLUTION

Conflicting Trade-offs: Fast deterministic response time directly conflicts with multi-user throughput. General-purpose time-sharing schedulers introduce unbounded queueing delays during context switches and cache thrashing, rendering them incapable of guaranteeing the 10 ms alarm threshold.

Application of OS Evolution: Early *batch systems* optimized raw throughput at the expense of interactive response. *Time-sharing systems* introduced interactive multi-user scheduling but introduced variable latency. Modern *handheld OSs* prioritize aggressive power state transitions (e.g., deep sleep states) which clash with high-performance server throughput demands.

3. ALTERNATIVE CONCEPT & COMPLEXITY ANALYSIS

> **Alternative Scenario: Deployment of a Single General-Purpose OS Across All Hospital Devices**
>
> If a standard general-purpose OS (e.g., standard Linux kernel with `Completely Fair Scheduler - CFS`) were deployed on bedside monitors:
>
> - **Worst-Case Alarm Delay:** Scheduling latency increases from bounded real-time upper limit (`O(1)` execution, < 2 ms) to unbounded worst-case delays (`O(N)` task contention, potentially exceeding 200 ms under heavy load), violating the 10 ms requirement.
> - **Memory Footprint:** Increases from ~4 MB (microkernel/RTOS) to 2 GB+ per monitor due to background daemons and graphic stacks.
> - **Space/Time Complexity Comparison:** RTOS deterministic lookup `O(1)` time and `O(1)` static memory vs. General Purpose OS `O(log N)` queue insertion time and dynamic memory overhead `O(M)`.